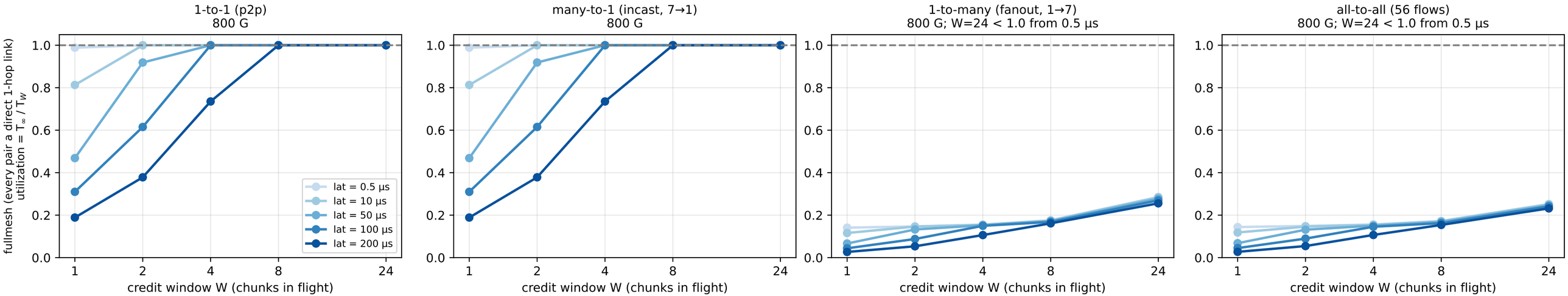
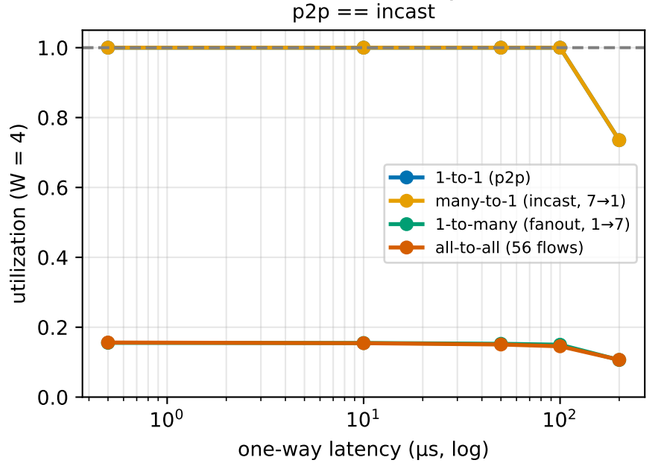
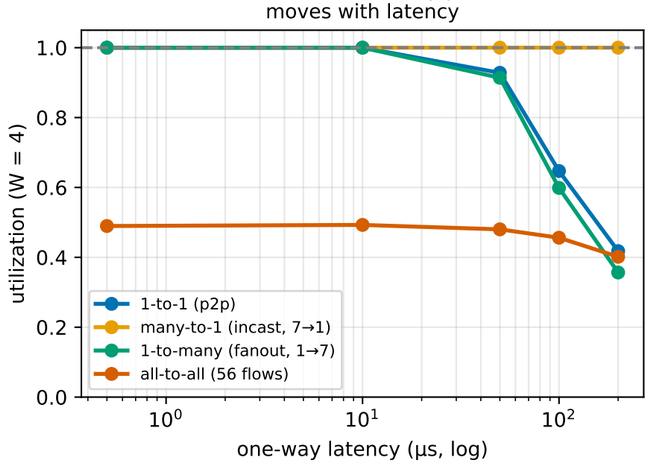
Credit flow control across traffic patterns at FIXED bandwidth (800 G, 8-GPU Su fabric, 256 MiB per flow):
utilization vs credit window W and one-way latency — dedicated (p2p/incast) vs shared (fanout/alltoall) windows, fullmesh vs switched







Utilization at 800 G, W = 4

lat(µs)	0.5	10	50	100	200
--- fullmesh ---					
p2p	1.00	1.00	1.00	1.00	0.74
incast	1.00	1.00	1.00	1.00	0.74
fanout	0.16	0.15	0.15	0.15	0.11
alltoall	0.16	0.15	0.15	0.15	0.11
--- switched ---					
p2p	1.00	1.00	0.93	0.65	0.42
incast	1.00	1.00	1.00	1.00	1.00
fanout	1.00	1.00	0.91	0.60	0.36
alltoall	0.49	0.49	0.48	0.46	0.40

W needed for full utilization shifts right ~linearly with latency (credit RTT); fanout/alltoall need ~W×flows credits (shared 1-VC window). Fullmesh T∞ is 7 links wide → fanout/alltoall capped even at W=24 (0.26/0.23).

